

Rajat Verma

rajatverma212@gmail.com | +91-8514944176 | throw-away-qq.github.io

Professional Summary

Lead Data Scientist with 8+ years of experience architecting production ML systems, reusable frameworks, and scalable pipelines on Google Cloud Platform. Built a YAML-driven Python framework at The Home Depot adopted by 10+ data scientists, reducing deep learning experimentation from months to under one week. Led a team of 8 to deliver Next Best Action deep learning recommendation models. Specialist in NLP, recommendation systems, causal inference, and MLOps with demonstrated business impact: 15.3% CTR improvement, 3.7% incremental lift, and \$150K+ in quantified savings.

Programming Skills

- **ML/DL:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, FAISS, HuggingFace Transformers
- **NLP/GenAI:** LLMs (Gemini, Gemma, Pegasus), Fine-tuning, RAG, Prompt Engineering, LangChain, Agentic AI, Multi-Agent Systems
- **Cloud & MLOps:** GCP (Vertex AI Pipelines, BigQuery, Cloud Composer, GCS), Apache Airflow, Docker, MLflow, CI/CD
- **Data & Languages:** Python, SQL, BigQuery, Pandas, PySpark, Spark, R, Flask, FastAPI, Streamlit, Git

Work Experience

The Home Depot (via Decision Culture)

Remote, India

Lead Data Scientist

February 2022 – Present

Tech Stack: Python, GCP (Vertex AI Pipelines, BigQuery, Cloud Composer), Apache Airflow, PyTorch, FAISS, MLflow

- Recognized with the **Orange Hand Award (x2)** for outstanding individual contribution.
- Architected a YAML-driven Python framework that orchestrates the full ML lifecycle (feature generation, model training, and batch scoring) across GCP services including BigQuery, Vertex AI Pipelines, and Cloud Composer. Reduced deep learning experimentation from months to under one week. Adopted by 10+ data scientists, powering 10+ models with 3 in production testing. Currently integrating MLflow for experiment tracking and model registry.
- Spearheaded Home Depot's transition from siloed per-class XGBoost models to a standardized deep learning stack (LSTM and Transformer), with every migrated model outperforming its predecessor, establishing a scalable, repeatable paradigm for recommendation modeling across the organization.
- Built a reusable causal inference framework on GCP for quantitative decision-making where A/B testing is infeasible, including per-customer revenue impact analysis of email opt-outs. Validated at greater than 99% alignment against a live A/B test, earning adoption across cross-functional teams for informed retention strategies.
- Developed a FAISS-based Lookalike Model on Vertex AI to identify high-value prospect customers, delivering 3.7% incremental lift over the existing approach with batch inference orchestrated via Cloud Composer.
- Built an automated BigQuery SQL pipeline using reusable templates for feature engineering, generating TensorFlow-compatible training datasets and eliminating manual query writing across the team.
- Conducted an internal workshop on Agentic AI Workflow Fundamentals for data science peers, building and open-sourcing a multi-agent Natural Language to SQL system as educational material. Demonstrated secure multi-LLM orchestration (Gemini, Gemma via OpenRouter), guardrail-based input validation, schema introspection, and Streamlit deployment, upskilling the team on production-grade AI agent architecture.
- Mentor and upskill data scientists on deep learning training diagnostics, production code standards, and architectural best practices. Lead code reviews and oversee integration quality across the team repository.

Times Internet

Delhi, India

Data Scientist

December 2020 – February 2022

- Deployed a personalized news recommendation pipeline serving 1.5M Economic Times subscribers, lifting click-through rate from 4.8% to 15.3%. Built end-to-end orchestration via Apache Airflow for scheduling,

monitoring, and logging.

- Fine-tuned Google's Pegasus for financial news summarization on ET Markets data, an early production LLM application, and shipped it as a Flask API currently serving live traffic.
- Designed a two-stage keyword recommendation system for editorial articles: BM25 retrieval over Wikipedia-indexed Solr followed by BERT embedding reranking, improving content tagging and discoverability.
- Hack and Hustle 3.0 Finalist (2022): Led a 3-person team to build a real-time Twitter social listening API delivering per-stock sentiment, subjectivity, and engagement analytics.

ZS

Pune, India

Data Science Associate

July 2018 – December 2020

- Automated rare disease identification from Electronic Health Records using a GAN-based PU-Classifer, compressing a multi-week manual process to 10 work hours.
- Built a resource-level compliance prediction model for a Fortune 100 Hi-Tech client. Proactive interventions driven by the model reduced non-compliance by 23% and saved \$150K within 6 months.
- Modeled asymptomatic NASH progression from EHR data (86% AUROC), identifying 4 novel non-invasive markers for fast-progressing patients. Adopted for clinical trial enrollment prioritization and focused treatment.
- Mined unstructured clinical trial text to surface competitive landscape insights and research trend analysis for therapeutic area strategy.

ZS

Pune, India

Data Science Intern

January 2017 – June 2017

- Built a multi-class sentiment analysis pipeline on noisy social listening data, benchmarking bidirectional LSTM, TF-IDF Random Forest, and Word2Vec MaxEnt, achieving 72% accuracy and 0.45 Cohen's Kappa.

Education

Indian Institute of Technology Kharagpur

West Bengal, India

Dual Degree (B.Tech and M.Tech) in Industrial Engineering

April 2013 – March 2018

Certifications

- **Generative AI Leader Professional Certificate** – Google Cloud (Coursera), October 2025 (Verify Credential)

Projects

- **Multi-Agent SQL Generator:** Developed an end-to-end NLP system orchestrating multiple LLMs (Gemini/Gemma) for query classification, schema introspection, and SQL generation. Integrated strict validation guardrails to prevent SQL injection and deployed via a Streamlit interface.
- **Screen Scape** (throw-away-qq.github.io/ScreenScape): Client-side web tool using React, Vite, and Tailwind CSS with daisyUI, enabling users to visualize and compare computer monitor sizes interactively in real-world scale.
- **Sequential Recommender in Production – RecBole** (throw-away-qq.github.io/recbole_to_production): Comprehensive guide and implementation for deploying RecBole's Sequential Models (SHAN, SINE) as production-ready FastAPI endpoints, bypassing RecBole's experimentation-focused interfaces to serve recommendations using only item history without user IDs.
- **Motion Following Camera:** Built using Arduino Uno and IR sensors for autonomous object tracking.